

MA661E	Exploratory Data Analysis, visualization and storytelling
Data analysis	
Lecture	Contents
1	Review the following concepts in linear algebra Vector: Length, inner Product, Angle, Projection, Rotation Matrix: Multiplication, Determinant, Inverse, Orthogonal matrices Similarity, Dissimilarity Measures (Appendix A pp. 418 - 423)
2	Chapter 1 , Introduction (pp. 3 -24) Some examples of data: Text document, Lung cancer, Leukemia Oronsay, Iris Preliminary treatment of data: Centerlization, Normalization Z-score, Sphering
3	Chapter 2 , Dimension Reduction(Linear): sections 1 - 5 (pp. 31- 56) Principal Component Analysis (PCA): Covariance approach Correlation approach Singular Value Decomposition (SVD) Non-negative Matrix factorization (NMF) Factor Analysis
4	Chapter 2 , Dimension Reduction: sections 6 - 8 (pp. 56 - 73) Linear Discreminant Analysis (LDA) Random Projection Dimension estimate (global & local): Nearest neighbor, Correlation, Maximum Likelihood, Parking Numbers
5	Chapter 3 , Dimension Reduction(nonlinear): sections 1~2 (pp. 79-108) Dimension reduction: Nonlinear approach Multidimensional scaling (MDS): Classical MDS Metric MDS (Scaling by Majorizing a Complicated Fucntion) Non-metric MDS Manifold Learning: Embedding, Isometric Feature Mapping (ISOMAP), Hessian Egenmaps
6	Chapter 5 , Finding Clusters: sectiona 1 - 3 (pp.155 - 167) Hierarchical methods: Single, Complete, Average, Centroid Linkage, Ward' method; Dendrogram, k-Means
7	Chapter 5, Finding Clusters: sections 4 - 6 (pp.167 - 200) Spectral, None-negative Matrix Factorization, Minimum Spanning trees Clustering; Rand index, Gap statistics.
8	Chapter 6 , Model Based Clustering: sections 1- 5 (pp. 205 - 230) Finite Mixture, Multivariant Mixture, Expectation Maximization Hierarchical Agglomerative MBC Bayes' Information Criterion BIC

9	Chapter 8 , Visualizing Clusters: sections 1-3 (pp. 301 - 312) Dendrogram, Treemaps, Rectangle plots, Data Image
Lab	Visualiztion Chapter 2
Seminar	You may choose one of the following sections for individuel study Chapter 3 , section 3; or any section in Chapter 4, Chapter 6 , section 6; Chapter 10, section 2 or a journal article on some topic which you are specially interested after being approved by the teacher. The oral presentation (about 10 min) should be prepared.
Remark	For lab and seminar, you may work in a team of max two persons

Week	Weekly activity	Suggested exercises
3	Chapter 1	1.2, 1.3, 1.6, 1.8
4	Chapter 2	2.1, 2.2, 2.3, 2.5, 2.8, 2.10 2.13, 2.19, 2.20, 2.28
5	Chapter 3 Chapter 5	3.1, 3.2, 3.10, 3.11, 3.13 5.1, 5.2, 5.3, 5.5
6	Chapter 5 Chapter 6	5.20, 5.22, 5.23, 5.25 6.2, 6.3, 6.6
7	Chapter 6	6.7 6.8, 6.12,6.17, 6.18
8	Chapter 8	8.1, 8.2, 8.4, 8.5, 8.7, 8.12

Examination	consists of Lab report, Seminar presentation, and written take-home exam
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